

A 3.2 Tbps 2.5D Integrated Silicon Photonic Transmitter for Co-Packaged Optics

Shuxiao Wang⁽¹⁾, Dehua Dai^{(1),(2)}, Huanwen Zhang^{(1),(2)}, Kai Zhang⁽¹⁾, Jingyu Wei^{(1),(2)}, Han Lu⁽¹⁾, Jianmin Zhang^{(1),(2)}, Dongyu Zhang^{(1),(2)}, Kelaiti Xiao⁽¹⁾, Yan Cai^{(1),(3),*}

⁽¹⁾ State Key Laboratory of Materials for Integrated Circuits, Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, 865 Changning Road, Shanghai 200050, China, * yan.cai@mail.sim.ac.cn

⁽²⁾ University of the Chinese Academy of Sciences, Yuquan Road 19, Beijing, 100049, China

⁽³⁾ Shanghai Industrial μ Technology Research Institute, Shanghai 201800, China

Abstract We demonstrate a 3.2 Tbps 2.5D integrated silicon photonic transmitter supporting a data transmission of 200 Gbps PAM4 modulation per channel. Full-link operation is experimentally validated, achieving 4.01 pJ/bit energy efficiency at a total power consumption of 12.83 W, highlighting its potential for co-packaged optics. ©2026 The Author(s)

Introduction

The rapid growth of data center traffic driven by cloud computing and artificial intelligence is pushing interconnect bandwidth toward multi-terabit-per-second regimes, where conventional electrical interconnects and pluggable optical modules face fundamental limitations in bandwidth density, power consumption, and signal integrity [1-3]. Co-packaged optics (CPO) has therefore emerged as a promising solution by reducing electrical interconnect length and enabling higher integration density with improved energy efficiency [4,5].

Recent silicon photonic systems have advanced toward high-bandwidth optical interconnects through diverse integration approaches. A 1.6 Tbps silicon photonic engine based on fan-out wafer-level packaging (FOWLP) was demonstrated, achieving high

aggregate data rate via electronic–photonic co-integration [5]. In parallel, wavelength-division multiplexed transmitters have enabled Tbps-class operation, with an 8×200 Gbps integrated transmitter achieving a total capacity of 1.6 Tbps [6]. Furthermore, monolithically integrated silicon photonics has reached Tbps-class performance, as demonstrated by a single-chip 1.024 Tb/s PAM4 receiver implemented using CMOS-photonics co-integration [7]. However, scaling aggregate bandwidth beyond the current 1.6 Tbps level while maintaining compact integration and system-level efficiency remains a key challenge.

In this paper, we present a 3.2 Tbps 2.5D integrated silicon photonic transmitter supporting a data transmission of 200 Gb/s PAM4 per channel. The architecture integrates photonic integrated circuits (PICs) and electronic integrated circuits (EICs) on a silicon interposer via fine-pitch micro-bump bonding, with vertical interconnections enabled by through-silicon vias (TSVs) based routing. The transmitter achieves full-link high-quality transmission across 16 channels, with a total power consumption of 12.83 W and an energy efficiency of 4.01 pJ/bit. The proposed architecture provides a scalable and energy-efficient solution for next-generation co-packaged optical interconnect systems.

Silicon Photonic Transmitter Architecture and Assembly

The proposed 2.5D integrated silicon photonic transmitter is presented in Fig. 1(a). The PIC and EIC chips are flip-chip bonded onto the silicon interposer through micro-bumps, which is further connected to the underlying high-speed substrate via vertical TSVs and solder balls. The grating couplers on the inverted PIC is packaged by coupling with a fiber array (FA), which is fixed by a ceramic plate placed above the EIC and PIC. Fig. 1(b) shows a photograph

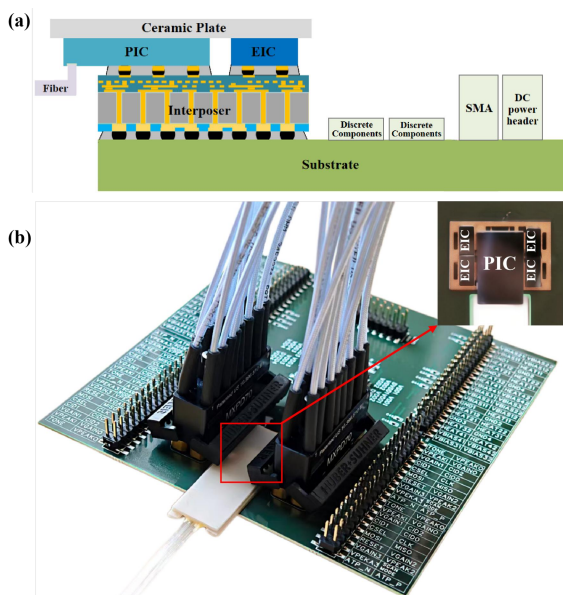


Fig. 1: A 2.5D integrated silicon photonic transmitter: (a) cross-sectional schematic, (b) photograph

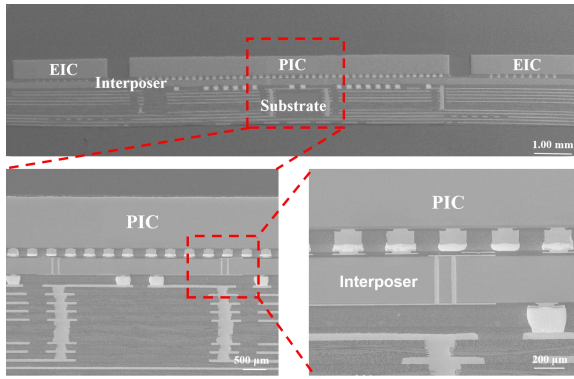


Fig. 2: Cross-sectional SEM of the 2.5D integrated silicon photonic transmitter with zoomed-in views.

of the integrated transmitter on a high speed substrate, with a magnified view presenting the inverted EICs and PIC. A groove is formed on the high-speed substrate to facilitate optical coupling and packaging. The PIC is fabricated on an 8-inch SiPh platform in a commercial foundry, and integrates 16-channel parallel differential Mach-Zehnder modulators (MZMs) and 32 grating couplers. The Cu/Ni/SnAg bumps on the PIC are fabricated using a wafer-level lithography and electroplating process. The commercial EIC with bumps is a high performance quad channel differential input and differential output linear open collector modulator driver. The bump pitch on both the EIC and PIC is 125 μm and the thickness is 65 μm .

The silicon interposer consists of through silicon vias (TSVs), front-side redistribution layers (RDLs), front-side RDL pads, back-side RDLs and backside bumps. The Bosch etching process is adopted to realize the etching of silicon vias with dimensions of 10 $\times$ 130 μm . After copper electroplating, the TSVs exhibit continuous and defect-free filling. The minimum pitch between TSVs is 40 μm . On the front side of the interposer, two layers of 3 μm -thick Cu RDL and 6.05 μm -thick Cu/Ni/Au pads are formed by electroplating. The back-side process involves temporary bonding, TSV back-side exposure, back-side redistribution and solder ball fabrication, and de-bonding. The thickness of the back-side RDL is 3 μm . The backside Cu/Ni/SnAg bumps have a thickness of 80 μm and a pitch of 180 μm . Finally, the front-side and back-side wiring of the silicon interposer are connected through TSVs, with fully filled and defect-free copper in the TSVs.

The high-speed substrate adopts a 10-layer organic stack structure. Impedance matching and loss control are achieved by optimizing the parameters of transmission lines and vias. High-speed signal layers and low-speed control layers are routed separately to reduce noise coupling. Discrete components such as

capacitors and resistors as well as direct current (DC) connectors are soldered onto the high-speed substrate using surface mount technology (SMT). Subsequently, the silicon interposer is flip-chip bonded to the high-speed substrate with underfills. One PIC chips and four EIC chips are then flip-chip bonded to the silicon interposer, with underfills. Finally, optical packaging using the FA are accomplished. Fig. 2 shows the cross-sectional SEM of the 2.5D integrated silicon photonic transmitter. It is observed that the TSV copper fill is fully dense with no visible voids or other filling defects, and the bump bonding interconnections are sound without open circuits.

Measurement setup and results

The measurement setup used to characterize the 2.5D silicon photonic transmitter is presented in Fig. 3(a). An O-band CW laser (1310 nm) is coupled into the PIC through a polarization controller. The PAM4 electrical signals are generated by an arbitrary waveform generator (AWG) and transmitted through the silicon interposer to the driver circuit. The modulated optical signal is then amplified and characterized using a high-speed oscilloscope. A 4-tap feed-forward equalizer (FFE) is applied, and the input data is a random code with a code length of 1024.

Fig. 3(b) presents the measured eye diagrams of 16 channels operating at 200 Gbps PAM4. Clear and well-defined four-level eyes are observed across all channels, indicating good linearity and bandwidth performance of the modulator-driver co-design. The extracted extinction ratio (ER) ranges from approximately 1.792 dB to 2.264 dB across the 16 channels. The average ER is 2.06 dB, demonstrating good channel-to-channel consistency enabled by the 2.5D integration scheme. A slight degradation can be attributed to electrical bandwidth limitation, RF routing loss on the interposer, and possible impedance mismatch at higher frequencies. The measured bit error rate (BER) performance of one channel as a function of data rate is shown in Fig. 3(c). As the data rate increases from 100 Gbps to 200 Gbps, the BER gradually degrades due to increased inter-symbol interference (ISI), bandwidth limitation, and noise accumulation. At 200 Gbps, the BER is on the order of 10^{-2} , which indicates that reliable transmission can be achieved with stronger forward error correction (FEC) schemes.

The total power consumption of the transmitter is composed of three independently supplied components: a 3.3 V supply for the

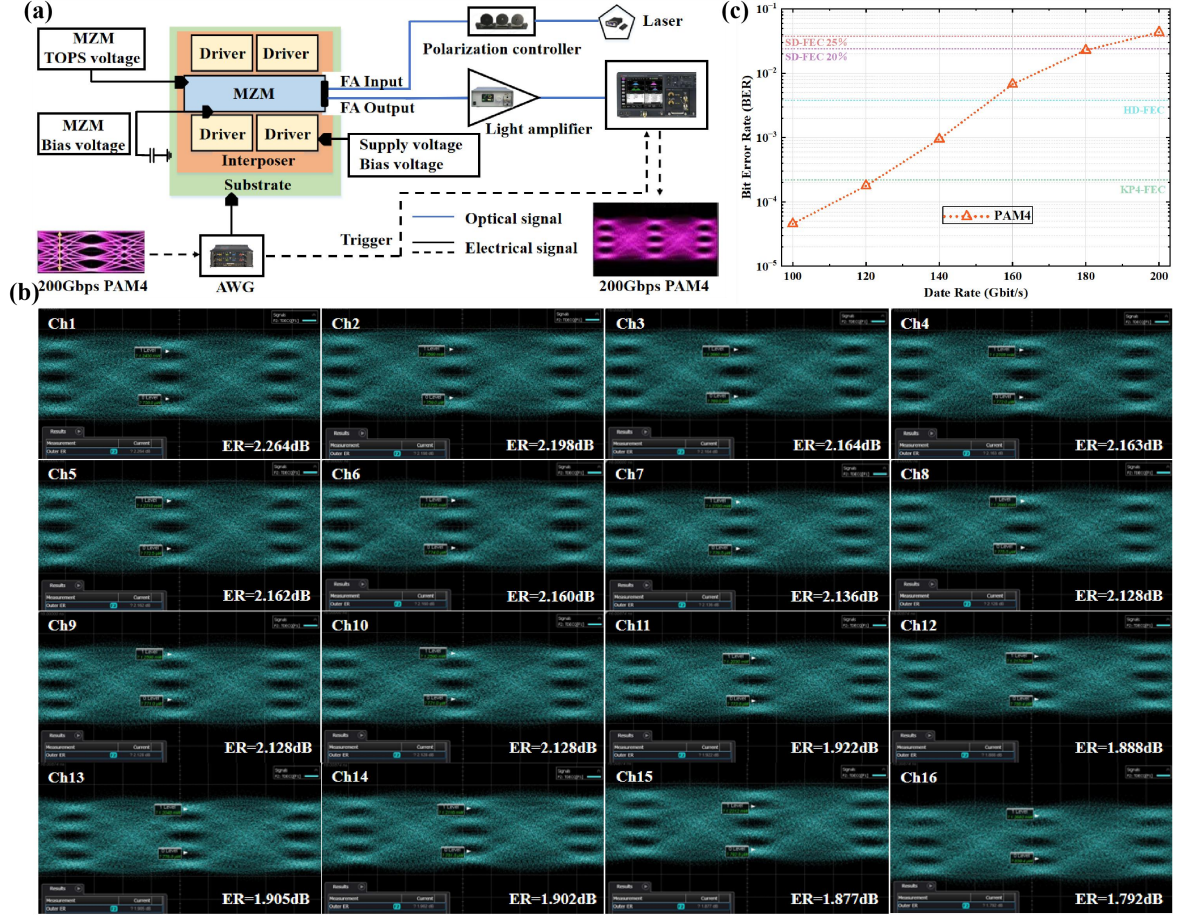


Fig. 3: (a) Eye diagram measurement setup for PAM4 transmission. (b) Measured eye diagrams of 16 channels operating at 200 Gbps PAM4. (c) Measured BER versus data rate for a single channel under PAM4 modulation.

digital and analog circuits of the EIC, a shared 5 V supply for the open-collector output stage of the EIC and the bias voltage of the PIC modulators, and a 1.5 V supply for per-channel thermal tuning of the PIC with 16 channels in total.

The power consumption of the EIC supply, the open-collector and modulator bias branch, and the thermal tuning of the PIC are calculated as follows:

$$P_{Elec} = V_{Elec} \times I_{Elec} \quad (1)$$

$$= 3.3 \text{ V} \times 1.56 \text{ A} = 5.148 \text{ W}$$

$$P_{Bias} = V_{Bias} \times I_{Bias} \quad (2)$$

$$= 5 \text{ V} \times 1.44 \text{ A} = 7.2 \text{ W}$$

$$P_{Hot} = V_{Hot} \times I_{Hot} \times N \quad (3)$$

$$= 1.5 \text{ V} \times 0.02 \text{ A} \times 16$$

$$= 0.48 \text{ W}$$

The total system power consumption and energy efficiency are given as follows:

$$P_{Total} = 5.148 \text{ W} + 7.2 \text{ W} + 0.48 \text{ W} \quad (4)$$

$$= 12.828 \text{ W}$$

$$EE = 12.828 \text{ W} / 3.2 \text{ Tbps} = 4.01 \text{ pJ/bit} \quad (5)$$

Under operation at $16 \times 200 \text{ Gbps PAM4}$,

corresponding to an aggregate data rate of 3.2 Tbps, the assembled 2.5D integrated silicon photonic transmitter exhibits a total power consumption of 12.83 W, resulting in an energy efficiency of 4.01 pJ/bit. Among the total power consumption, the 5 V bias supply accounts for 56.1%, representing the dominant contribution, while the EIC core power accounts for 40.1%, and the thermal tuning power contributes only 3.8%. The proposed 2.5D integrated silicon photonic transmitter achieves competitive energy efficiency at multi-terabit data rates, demonstrating its suitability for high-performance CPO applications.

Conclusions

In this paper we proposed and demonstrated a 3.2 Tbps 2.5D integrated silicon photonic transmitter based on a silicon interposer with TSVs. The transmitter has been verified to support single-channel 200 Gbps PAM4 eye-diagram transmission with a total of 16 channels, achieving a transmitter energy efficiency of 4.01 pJ/bit. This 2.5D integrated silicon photonic transmitter serves as a key enabling solution for next-generation high-performance, low-power optical interconnect systems.

Acknowledgements

This work is supported by the National Natural Science Foundation of China (No. 62405348, No. 92473104, No.62474180).

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